SPECIAL COMMUNICATIONS
Methodological Advances

From Total Volume to Sequence Maps: Sophisticated Accelerometer Data Analysis

MAI J. CHINAPAW[1], XINHUI WANG[1,2], LARS BO ANDERSEN[3], and TEATSKE M. ALTENBURG[1]

[1]*Department of Public and Occupational Health and Amsterdam Public Health Research Institute, VU University Medical Center, Amsterdam, THE NETHERLANDS;* [2]*College of Computer Science, Qinghai Normal University, Xining, Qinghai, CHINA; and* [3]*Faculty of Education, Arts and Sports, Western Norway University of Applied Sciences, Campus Sogndal, NORWAY*

TABLE 1. Symbol definitions based on the accumulation of SB and PA intensity.

Symbol	Bout Length (min)	Cut Point	Tolerance (%)
1	≥30	SB <100 counts per minute	0
2	10–29.9	SB	0
3	0–9.9	SB	0
4	0–9.9	LPA ≥25 cp15s	0
5	10–29.9	LPA	10[a]
6	≥30	LPA	10[a]
7	0–4.9	MPA ≥574 cp15s	0
8	5–9.9	MPA	10[a]
9	≥10	MPA	10[a]
10	0–4.9	VPA ≥1003 cp15s	0
11	5–9.9	VPA	10[a]
12	≥10	VPA	10[a]

cp15s = counts per 15 s.
[a]The lower threshold tolerance was restricted to an absolute tolerance of three consecutive minutes to prevent too much time below the specified activity level within a bout.

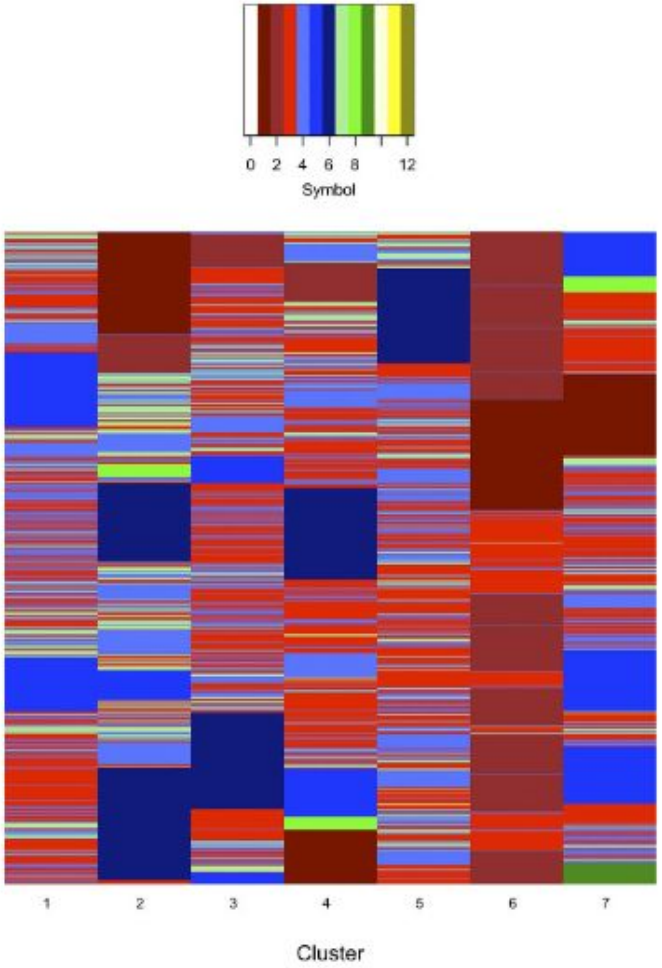


FIGURE 1—Example of a sequence map of one child in each cluster.